Laser Physics Letters

Astro Ltd.

LETTER

Two-dimensional photoacoustic imaging of femtosecond filament in water

To cite this article: F V Potemkin *et al* 2018 *Laser Phys. Lett.* **15** 075403

View the article online for updates and enhancements.

Related content

- Laser control of filament-induced shock wave in water
  F V Potemkin, E I Mareev, A A Podshivalov et al.
- Highly extended high density filaments in tight focusing geometry in water: from femtoseconds to microseconds
  F V Potemkin, E I Mareev, A A Podshivalov et al.
- Ultrashort filaments of light in weakly ionized, optically transparent media
  L Bergé, S Skupin, R Nuter et al.

This content was downloaded from IP address 129.8.242.67 on 03/06/2018 at 12:45

IOP Publishing | Astro Ltd Laser Physics Letters
Laser Phys. Lett. **15** (2018) 075403 (5pp) https://doi.org/10.1088/1612-202X/aabc99

Letter

Two-dimensional photoacoustic imaging of femtosecond filament in water

F V Potemkin[1,2], E I Mareev[1,2], B V Rumiantsev[1,2], A S Bychkov[1,3], A A Karabutov[2,3], E B Cherepetskaya[1,3] and V A Makarov[1,2,3]

[1] Faculty of Physics, M.V. Lomonosov Moscow State University, Leninskie Gory bld.1/2, Moscow 119991, Russia
[2] International Laser Center, M.V. Lomonosov Moscow State University, Leninskie Gory bld.1/62, Moscow 119991, Russia
[3] The National University of Science and Technology MISiS, Leninski Prospect 6, Moscow 119991, Russia

E-mail: potemkin@physics.msu.ru

Received 23 March 2018, revised 1 April 2018
Accepted for publication 7 April 2018
Published 30 May 2018

Abstract

We report a first-of-its-kind optoacoustic tomography of a femtosecond filament in water. Using a broadband (~100 MHz) piezoelectric transducer and a back-projection reconstruction technique, a single filament profile was retrieved. Obtained pressure distribution induced by the femtosecond filament allowed us to identify the size of the core and the energy reservoir with spatial resolution better than 10 μm. The photoacoustic imaging provides direct measurements of the energy deposition into the medium under filamentation of ultrashort laser pulses that cannot be obtained by existing techniques. In combination with a relative simplicity and high accuracy, photoacoustic imaging can be considered as a breakthrough instrument for filamentation investigation.

Keywords: optoacoustic tomography, femtosecond filament, photoacoustic imaging, back-projection reconstruction

(Some figures may appear in colour only in the online journal)

1. Introduction

Ultrashort laser pulse filamentation is one of the hot topics in modern photonics. Propagation of a high-intensity ultrashort laser pulse in a bulk material leads to a nonlinear laser-matter interaction that is governed by a dynamical balancing of self-focusing and plasma defocusing [1]. The laser pulse energy transport along the propagation distance can be achieved by a control over the laser-induced plasma generation in the filament. Thus, the evolution of the laser pulse and the laser-induced electron density distribution in the filament are of fundamental interest. Prior estimates of such parameters have been obtained through numerical calculations and several indirect experimental methods including optical probing [2–5], plasma fluorescence [6], shadow photography [7, 8], sonography [9], interferometry [10], polarimetry [11] and secondary electrical discharges [12].

These experimental methods are focused on the measurements of the plasma column that limits the detection capability and the filament core with sufficiently high intensity; and electron density can be detected. The laser pulse energy reservoir that surrounds the high-intensity core of the filament actually drives the filamentation process, but cannot be registered due to its low intensity.

In this letter, we present a novel photoacoustic (PA) imaging method to measure a single filament profile along the path of the filamenting pulse in a condensed transparent dielectric. Particularly, a water sample was used in the experiment. This method allows us to identify clearly the location of the core and the energy reservoir of the filament with a micron spatial

 © 2018 Astro Ltd Printed in the UK

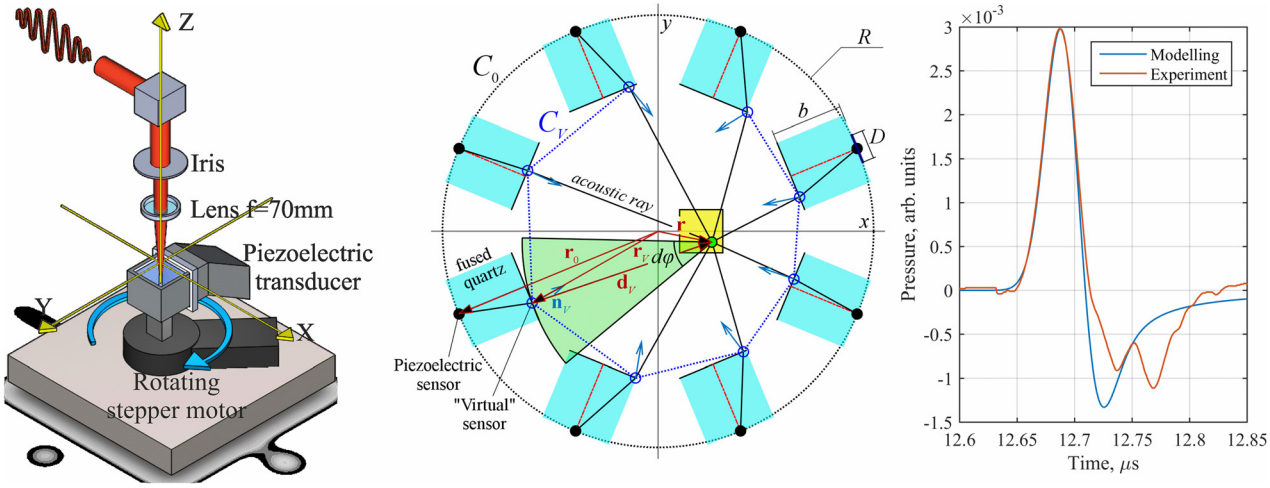

Figure 1. The schematic of the experimental setup (left). 'Actual' (C_0) and 'virtual' (C_V) detection curves for 8 sensor positions (center). The characteristic experimental and numerically modelled PA signal from the filament (right).

resolution that cannot be obtained by existing techniques and is highly demanded both for fundamental science and applications.

2. Experimental setup

In our experiments, a Cr:Forsterite laser system was applied ($\lambda = 1240$ nm, pulse duration ~150 fs, laser energy up to 2 mJ with fluctuations less than 5%, intensity contrast about ASE, and repetition rate of 10 Hz). The laser pulse energy was varied by a half wave plate and a Glan prism. The diameter of the laser beam was truncated to 2.9 mm by the iris. The Ge detector (Thorlabs PDA50-B) measured the incoming laser energy. The laser beam with 30 μJ energy was focused into the water cell (at open water-air interface) by a lens with a focusing distance of f = 70 mm, which leads to single filament generation accompanied by a bright supercontinuum and conical emission. The water cell was connected to a broadband (~100 MHz) piezoelectric transducer and mounted on a rotating stepper motor and an automated three-axis micromanipulator (figure 1). The acoustic signals were acquired by the oscilloscope (AKIP-4119), and averaged over 16 data points (at each angle position). A three-axis micromanipulator allowed for the shifting of the filament position with a micron spatial resolution. The experimental setup was fully automated.

3. Results and discussion

Filamentation in transparent materials deposits a significant energy delivery into the medium and generates cavitation bubbles and shock waves. The maximal energy delivery into mechanical post-effects is achieved in the superfilamentation regime (it can reach 15%) [8]; however a single filament emits acoustic waves that can be registered confidently. As noted in [13], at a density of the absorbed laser energy, recombination of the filament plasma and further thermalization results in a formation of a thermal source, which thermoelastically generates acoustic waves [14, 15]. Acoustic nonlinearity does not distort the temporal form of the pressure pulse significantly. The typical signal from the piezoelectric sensor has an asymmetric N-shape [16], corresponding to a steep positive pressure peak followed by a negative pressure region. The duration of the laser pulse (~170 fs), the plasma recombination time (~ps) and the thermalization time (~ps) in water is much shorter than the characteristic propagation time of the acoustic wave $w_0/c \sim 4$ ns (where $w_0 \sim 5\,\mu$m is the characteristic filament radius, $c = 1477\,\mathrm{m\,s^{-1}}$ is the speed of sound in water) and the characteristic thermal conduction time $w_0^2/\chi \sim 0.2$ ms (where $\chi = 0.143\ \mathrm{mm^2\,s^{-1}}$ is the thermal diffusivity of water). Therefore, we can assume that the heat source was formed instantaneously, and the heat power density of the source is $H(\mathbf{r},t) = Q(\mathbf{r})\,\delta(t)$, where $Q(\mathbf{r})$ is the energy density released at the point determined by vector $\mathbf{r} = \{x, y\}$ (see figure 1) and $\delta(t)$ is the Dirac delta function of time t. Neglecting thermal conduction, acoustic attenuation and nonlinearity, we can write the wave equation for the acoustic pressure field $p(\boldsymbol{r},t)$ in the acoustically homogeneous medium (water):

$$\left(\partial^2/\partial t^2 - v_s^2\Delta_\perp\right) p(\mathbf{r},t) = p_0(\mathbf{r})\, d\delta(t)/dt, \qquad (1)$$

where $\Delta_\perp$ is the transverse component of the Laplace operator in the cylindrical coordinate system, $p_0(\mathbf{r}) = \left(\beta v_s^2/c_v\right) Q(\mathbf{r}) = p(\mathbf{r}, t = 0)$ is the initial pressure distribution in the medium, $\beta = -\rho^{-1}(\partial\rho/\partial T)_p$ is the volumetric thermal expansion coefficient, ρ and T are density and temperature, v_s is speed of sound and c_v is the specific heat capacity at a constant volume.

In a homogeneous medium, $p_0(\mathbf{r})$ can be reconstructed using a back-projection algorithm [17] (BPA) by integration over the detection surface. BPA has been successfully used in the numerical study of filamentation in air [18]. The ultrasonic pulses were recorded by a flat $LiNbO_3$ piezoelectric sensor with a diameter of $D = 6$ mm optically connected to a cylindrical prism made of a fused quartz of the height $b = 32.5$ mm (figure 1). The refraction of the ultrasonic pulse generated by the filament in water at the water-quartz boundary and its propagation in fused silica changed the propagation time and pulse amplitude. Therefore, it was necessary to

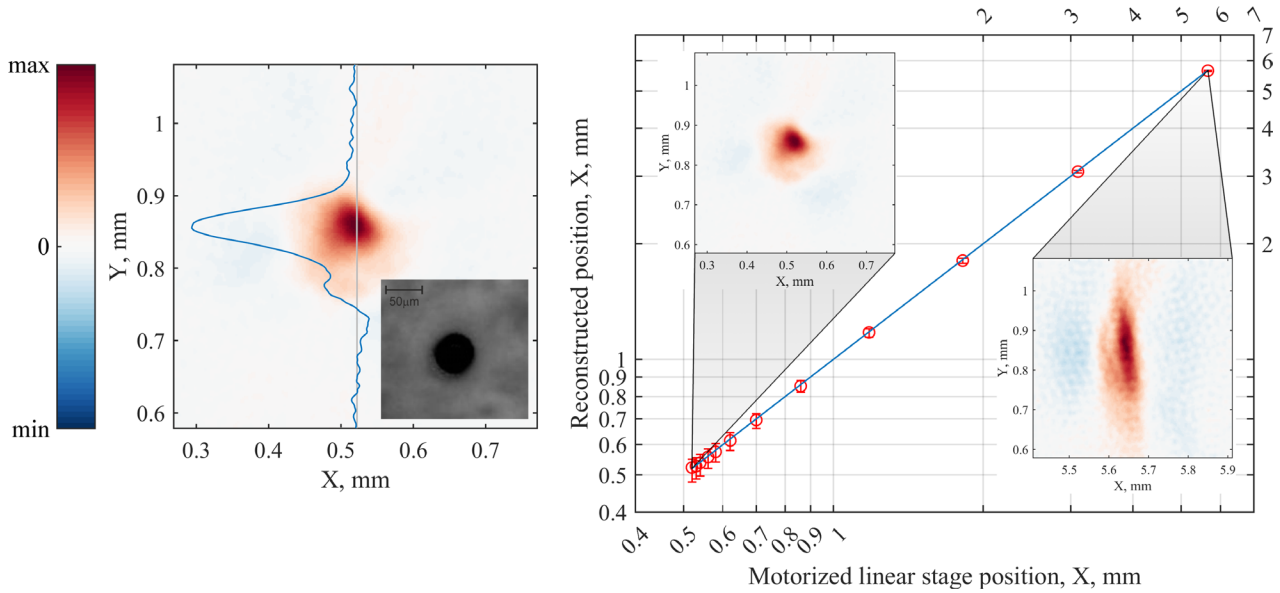

Figure 2. PA image of the filament closest to the center of the detection curve with its cross-sections through the point with maximum amplitude (left), the inset shows the impact of the filament on the photographic paper. The dependence of the filament position retrieved from the PA images on the actual filament shift (right), the insets show the PA image changes.

slightly modify the BPA used in [18] and integrate over some 'virtual' detection curve C_V (figure 1) that passed through the points of refraction ('virtual' sensors) of acoustic rays. C_V depended on the position of the image point determined by $\mathbf{r}$ and was not always a circle, so the BPA for arbitrary detection curves [19] was combined with the BPA for 2D-sources emitting cylindrical waves [20], and modified to account for refraction using geometrical acoustics. The reconstructed initial pressure distribution in the image point:

$$p_0^{(b)}(x,y) \simeq -\frac{2}{\Omega_V}\int_{C_V}\left(\int_{d_V/v_s}^{+\infty}\frac{1}{v_s t_a}\frac{\partial p(\mathbf{r}_0,t)}{\partial t}\bigg|_{t=t_a+t_b}\frac{dt_a}{\sqrt{v_s^2 t_a^2 - d_v^2}}\right)\frac{(\mathbf{n}_V\cdot\mathbf{d}_V)}{\varepsilon W}dC_V. \quad (2)$$

Here the acoustic ray emitted from the image point and recorded by the sensor at the point $\mathbf{r}_0$ undergo refraction at the point $\mathbf{r}_V$. t_a and t_b are the propagation times in water and in quartz, respectively. W is the pressure transmission coefficient for the water-quartz boundary at $\mathbf{r}_V$. ε is the ratio of the pressure amplitudes in quartz at $\mathbf{r}_V$ and at $\mathbf{r}_0$, $d_V = \mathbf{r} - \mathbf{r}_V$, $\mathbf{n}_V$ —unit normal vector at $\mathbf{r}_V$ to the 'virtual' detection curve C_V pointing inwards (see figure 1). Ω_V is the full angular aperture of C_V seen from the origin.

The filament-induced ultrasound was experimentally recorded at points of the 220° circular arc by rotating the sensor around the axis that was parallel to the filament axis with a 1° angular step (figure 1). The experimental signals were filtered by a bandpass filter (the response of frequency f is proportional to $\exp\left[-(f_l/f)^8 - (f_l/f)^2 - (f/f_h)^4\right]$, where $f_l = 1$ MHz, $f_h = 150$ MHz—lower and upper cutoff frequencies) to reduce noise. Figure 2 shows the reconstructed PA tomographic images (201 × 201 pixels, 0.5 mm × 0.5 mm). The faintly visible blue areas in the experimental images (the regions with 'negative energy deposition') emerge mainly due to the filtering of low frequency components of the experimental signals and should be neglected.

We carried out a series of experiments to investigate how the image of the filament changed with the distance from the filament to the center of the detection curve. The filament was translated along the x axis (figure 1) and the centers of the PA images were shifted so that they coincided with the expected filament locations. The location of the pixel with the maximum amplitude was considered a reconstructed position of the filament.

Figure 2 shows the dependence of the reconstructed filament x coordinate on the actual x coordinate calculated from the number of steps taken by the micromanipulator. The error bars show the x-positions of the half-maximum levels. The actual position of the filament is always well within the error bars. Therefore, PA imaging allows the position of the filament to be determined with a high accuracy (the minimal registered shift is 10 μm). This implies that the relative position of multiple separate filaments can be reconstructed with ~10 μm accuracy.

The full width at half maximum (FWHM) of the image along the x axis remains almost constant while the FWHM along y axis increases, which correlates with the results in [18, 21]. The angular size of the PA image as seen from the origin corresponds to the angular size of the sensor [21]. The resolution along the x axis is determined mainly by the effective receiving bandwidth [21]. The acoustic attenuation acts as a low-pass filter with the frequency response $\sim \exp\left[-\alpha f^2 d\right]$, where $\alpha = 25$ mm^{-1} GHz^{-2} is the acoustic attenuation coefficient of water and d is the distance from the filament to the ('virtual') receiver. Additionally, the acoustic wave with a finite radius of curvature is registered by a flat piezoelectric transducer, which results in blurring due to the integration of the signal over the surface of the transducer.

We conducted numerical modeling to quantify the blurring effects and estimate the actual FWHM of the plasma core

using the technique described in [18]. The model Gaussian heat sources, $Q(\mathbf{r}) \sim \exp\left(-r^2/w^2\right)$ with various radii w from 5 μm to 70 μm, were placed at the origin, and the signals on 32 identical sensors were calculated accounting for the refraction, flat shape and finite size of each sensor. The geometrical parameters of the detection circle (b, R, D) were taken from the experiment setup. The numerical 32-element array was a full circle with the angular aperture of 2π.

It was found that acoustic attenuation adds $5 \div 10$ μm and the integration over the flat surface of the transducer adds ~15 μm to the FWHM of the PA image of the Gaussian heat source. Altogether the experimental setup with the chosen geometrical parameters (b, R, D) adds roughly 20 μm to the image FWHM, so the details of the heat source finer than 20 μm cannot be reliably resolved. Having analyzed the dependence of the FWHM of the PA image of the Gaussian heat source on its actual FWHM, we can estimate the actual FWHM of the filament. The FWHM of the PA image of the filament closest to the origin averaged over 8 directions (65 ± 6 μm), corresponding to the actual FWHM of 45 ± 6 μm.

In order to prove the obtained size of the filament we applied a method based on the single-shot darkening of photographic plates [6, 22]. In this method, the filament impacts the photographic paper, which leads to the darkening of the paper in the core of the filament. The determined diameter of the filament averaged over 10 data points is 58 ± 6 μm, which also correlates with the literature data [23–25].

The halo around the core in the PA images can be interpreted as the energy reservoir. Most experimental techniques for filament investigation obtained information about the filament-induced plasma channel [6, 10, 26] and could not obtain any information about the reservoir, since the plasma is not generated in the reservoir itself. PA imaging deals with heat sources, thereby the reservoir could induce acoustic waves in the regions where no plasma is generated. To the best of our knowledge, the experimental investigations of the energy reservoir to date were limited by methods based on applying pinholes with different diameters [27].

4. Conclusion

In conclusion, we present a novel approach for the characterization of filament propagation in condensed matter. The photoacoustic imaging gives a unique opportunity to observe the energy delivery into the medium under femtosecond filamentation directly. This method gives information both about the energy reservoir and the filament core. The estimated size of the single filament core is about 50 μm, and the energy reservoir is about 250 μm. The resolution of photoacoustic imaging is determined by the bandwidth of the piezoelectric transducer and the absorption of the medium. We believe that resolution of the method can reach several micrometers in condensed matter. In water, location and size of the core and the energy reservoir of the single filament can be retrieved with a spatial resolution better than 10 μm.

Acknowledgments

This work was supported by the Russian Science Foundation (grant No. 17-72-20130). The image reconstruction algorithm was developed and numerical modelling was conducted under the funding of NUST 'MISiS' Competitiveness Program by the Ministry of Education and Science of the Russian Federation (K2-2017-003) and Russian Science Foundation (16-17-10181). E I Mareev has a scholarship of BASIS foundation (Russia).

References

[1] Couairon A and Mysyrowicz A 2007 *Phys. Rep.* **441** 47–189
[2] Thиberge F, Liu W, Simard P, Becker A and Chin S 2006 *Phys. Rev.* E **74** 036406
[3] Tzortzakis S, Prade B, Franco M and Mysyrowicz A 2000 *Opt. Commun.* **181** 123–7
[4] Matijošius A, Piskarskas R, Gaižauskas E, Dubietis A and Di Trapani P 2004 *Nonlinear Anal. Model. Control* **9** 259–70
[5] Faccio D, Lotti A, Matijosius A, Bragheri F, Degiorgio V, Couairon A and Di Trapani P 2009 *Opt. Express* **17** 8193–200
[6] Point G, Brelet Y, Houard A, Jukna V, Milián C, Carbonnel J, Liu Y, Couairon A and Mysyrowicz A 2014 *Phys. Rev. Lett.* **112** 223902
[7] Minardi S, Gopal A, Couairon A, Tamoasuskas G, Piskarskas R, Dubietis A and Di Trapani P 2009 *Opt. Lett.* **34** 3020–2
[8] Potemkin F V, Mareev E I, Podshivalov A A and Gordienko V M 2015 *New J. Phys.* **17** 53010
[9] Yu J, Mondelain D, Kasparian J, Salmon E, Geffroy S, Favre C, Boutou V and Wolf J-P 2003 *Appl. Opt.* **42** 7117–20
[10] Bodrov S, Bukin V, Tsarev M, Murzanev A, Garnov S, Aleksandrov N and Stepanov A 2011 *Opt. Express* **19** 6829
[11] Woodworth J R, Zameroski N D, Johnson D L, Sarkisov G S, Blickem J R, Van De Valde D M, Starbird R L and Wilkins F L 2005 *Laser Triggering of Water Switches in Terrawatt-Class Pulse Power Accelerators (Sandia Report SAND 2005-6990)* (Albuquerque, NM: Sandia National Laboratories)
[12] Ionin A A, Kudryashov S I, Mokrousova D V, Seleznev L V, Sinitsyn D V and Sunchugasheva E S 2014 *Laser Phys. Lett.* **11** 16002
[13] Jukna V *et al* 2016 *Phys. Rev.* E **93** 063106
[14] Potemkin F V, Mareev E I, Podshivalov A A and Gordienko V M 2014 *Laser Phys. Lett.* **11** 106001
[15] Potemkin F V and Mareev E I 2015 *Laser Phys. Lett.* **12** 15405
[16] Brelet Y, Jarnac A, Carbonnel J, André Y, Mysyrowicz A, Houard A, Fattaccioli D, Guillermin R and Sessarego J 2015 *J. Acoust. Soc. Am.* **137** EL288–92
[17] Xu M and Wang L V 2005 *Phys. Rev.* E **71** 016706
[18] Bychkov A S, Cherepetskaya E B, Karabutov A A and Makarov V A 2016 *Laser Phys. Lett.* **13** 85401
[19] Burgholzer P, Matt G J, Haltmeier M and Paltauf G 2007 *Phys. Rev.* E **75** 046706
[20] Burgholzer P, Bauer-Marschallinger J, Grün H, Haltmeier M and Paltauf G 2007 *Inverse Problem* **23**
[21] Xu M and Wang L V 2003 *Phys. Rev.* E **67** 056605
[22] Méchain G, D'Amico C, André Y B, Tzortzakis S, Franco M, Prade B, Mysyrowicz A, Couairon A, Salmon E and Sauerbrey R 2005 *Opt. Commun.* **247** 171–80

[23] Dubietis A, Tamosauskas G, Diomin I and Varanavicius A 2003 *Opt. Lett.* **28** 1269–71
[24] Kolesik M, Wright E M and Moloney J V 2004 *Phys. Rev. Lett.* **92** 15–8
[25] Jarnac A, Tamosauskas G, Majus D, Houard A, Mysyrowicz A, Couairon A and Dubietis A 2014 *Phys. Rev.* A **89** 33809
[26] Dergachev A A, Ionin A A, Kandidov V P, Mokrousova D V, Seleznev L V, Sinitsyn D V, Sunchugasheva E S, Shlenov S A and Shustikova A P 2014 *Quantum Electron.* **44** 1085–90
[27] Liu W, Théberge F, Arévalo E, Gravel J F, Becker A and Chin S L 2005 *Opt. Lett.* **30** 2602–4